

Skills Review

Exercise No. 6

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Grade:

Course/Section: CPE4B

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Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer6.asm as the filename.

aZ
bY
cX
dW
.
.
.
zA

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.code

start:
    mov ah, 02h
    mov cx, 26          ; 26 letters, A to Z

    mov bl, 5ah         ; bl = 'Z'
    mov dl, 61h         ; dl = 'a'
    mov dh, dl          ; dh = 'a'

back:
    int 21h             ; display uppercase (dl)
    mov dl, bl          ; load lowercase
    int 21h             ; display lowercase

    mov dl, 0dh
    int 21h
    mov dl, 0ah
    int 21h
```

```

    dec bl          ; next uppercase letter
    inc dh          ; next lowercase letter

    mov dl, dh
    loop back

    mov dl, 0dh
    int 21h
    mov dl, 0ah
    int 21h

    mov ah, 4ch
    int 21h

end start

```

Output:

```

dW
eV
fU
gT
hS
iR
jQ
kP
lO
mN
nM
oL
pK
qJ
rI
sH
tG
uF
vE
wD
xC
yB
zA

D:\>

```

***The program actually starts displaying from 'aZ' as intended, but because the DOS/command prompt window has a limited number of visible lines, the earlier output ('aZ' and 'bY') scrolled off the top of the screen once all 26 lines were printed. The screenshot only captured the last visible lines, which is why it appears to start at 'cX' instead of 'aZ', this is a display/scrolling limitation of the terminal window, not an error in the program's logic. ***